Genetics: Key Concepts

Biology study notes

Study snapshot

Build a clear vocabulary for inheritance and simple genetic crosses.

Key ideas

- A gene is a DNA sequence associated with a functional product or trait.
- Different versions of a gene are called alleles.
- Genotype describes allele combinations; phenotype describes observed traits.
- In a simple dominant-recessive model, a dominant allele can mask a recessive one.
- A Punnett square lists possible allele combinations from a cross.

Try the questions on page 2 before checking your work.

Genetics: Key Concepts

Quick practice

1. For genotype Aa, how many different alleles are present?
2. In a simple model, which genotype is heterozygous: AA, Aa, or aa?
3. Cross Aa x Aa. What fraction of offspring are expected to be aa?
4. What term describes an observable trait?

Answer notes

1) Two allele forms: A and a
2) Aa
3) 1/4
4) Phenotype

Review the method, then retry any question you missed.